

cAluldron: Technical Blueprint

<https://github.com/Ckcinnabar/cAluldron.git>

Project: cAluldron – AI-Powered Recipe Generator

a. Problem Context and Project Summary

The cAluldron project reimagines how home cooks approach their ingredients. By simply uploading a photo of an ingredient, users receive complete cooking plans—including personalized recipes, nutritional breakdowns, and illustrated guides. The project addresses the common problem of decision fatigue and food waste in home kitchens, offering an intelligent system that transforms raw inputs into culinary inspiration. It bridges computer vision, language modeling, and probabilistic reasoning to produce creative yet practical cooking solutions.

b. Dataset

The system leverages multiple open-source datasets to achieve multimodal understanding: Food-101 (for ingredient recognition), Recipe1M+ (for recipe text generation), and USDA FoodData Central (for nutritional estimation). Together, these datasets provide the necessary balance of visual, textual, and quantitative information to power the model ensemble.

Food-101 contains 101,000 labeled food images across 101 categories, providing diverse visual contexts for CNN-based ingredient recognition. Recipe1M+ contributes over one million structured recipes, including ingredient lists, instructions, and images—enabling transformer models to learn language patterns related to food preparation. The USDA dataset anchors the project with precise nutritional metrics, allowing Bayesian networks to infer caloric and macronutrient values.

Data challenges include domain mismatch between datasets (restaurant food vs. home ingredients), variance in labeling standards, and size estimation from single images. To mitigate this, synthetic data augmentation, domain adaptation techniques, and heuristic calibration will be implemented. Ethical considerations include respecting dataset licenses and ensuring all user photos remain processed locally for privacy compliance.

c. Planned Architecture

The architecture integrates vision, language, and probabilistic inference into a unified pipeline. The workflow follows this data flow: Photo Upload → CNN Recognition → Recipe Generation → Nutrition Estimation → Illustration Creation.

- Ingredient Recognition – EfficientNetV2-S (PyTorch) identifies primary ingredients from uploaded photos and estimates portion size. The CNN produces feature embeddings later used for cross-modal recipe retrieval.

- **Recipe Generation** – GPT-2 Medium, fine-tuned on Recipe1M+, generates five or more diverse recipes per ingredient. Outputs include structured data such as cooking time, difficulty, and cuisine type.
- **Cooking Refinement** – A BiLSTM model reorders and refines recipe steps, improving logical flow and timing accuracy.
- **Nutrition Estimation** – A Bayesian probabilistic graphical model (PGM) maps recognized ingredients to USDA nutrition data, propagating uncertainty to provide $\pm 20\%$ confidence intervals.
- **Illustration Generation** – A Stable Diffusion v1.5 + ControlNet pipeline generates line-art style visuals for each cooking step. This offers interpretable, low-latency guidance while maintaining aesthetic consistency.

The total end-to-end processing time is targeted under 15 seconds with GPU acceleration, enabling near-real-time interaction.

d. User Interface Plan

The user interface (UI) serves as the interactive bridge between technical models and human usability. Built using Gradio for rapid prototyping, it will later expand into a Streamlit or Flask app for deployment. The UI will allow users to upload an ingredient photo, specify dietary preferences (vegetarian, gluten-free, etc.), and view AI-generated recipes complete with illustrations, cooking times, and nutritional facts.

Users will interact through a clean, card-based layout displaying five recipe suggestions. Each card contains an image, recipe title, difficulty level, cuisine tag, and an expandable view for full instructions. Step-by-step cooking guides will feature line-art images generated by Stable Diffusion + ControlNet, providing visual clarity for complex techniques such as dicing, sautéing, or whisking. Interactive features include thumbs-up/down feedback for continuous learning, timing notifications for critical steps, and tooltips explaining technical cooking terms.

The primary UX goal is transparency—showing users how AI made each decision, from ingredient detection confidence to calorie estimation rationale. This builds trust while fostering engagement with the cooking process.

e. Innovation and Anticipated Challenges

The innovation of cAIuldron lies in its convergence of multimodal AI—vision, language, and generation—within a consumer application. Unlike traditional recipe recommenders, this system creates recipes from scratch using real-world visual inputs. It not only recognizes what is in a user's kitchen but also teaches how to cook it, step by step, using visual storytelling.

Technical challenges include achieving robust recognition under poor lighting or cluttered backgrounds, managing model latency for real-time inference, and preventing recipe

hallucinations that could result in unsafe recommendations. To mitigate these, the project employs mixed-precision inference, post-processing rule filters for recipe validation, and model distillation for lightweight deployment.

Another challenge lies in integrating large generative models (Stable Diffusion and GPT-2) within resource-limited environments. The team plans to support GPU inference while offering fallback CPU modes through quantized versions of the models. This balance ensures accessibility for educational and hobbyist users while maintaining scalability.

f. Implementation Timeline

Week	Focus	Expected Outcome
Oct 20 – 26	Data cleaning and baseline setup	Working data loader, dataset exploration notebooks, environment verified.
Oct 27 – Nov 2	Model training and UI scaffolding	EfficientNetV2 baseline trained; initial Gradio UI prototype functional.
Nov 3 – 9	Recipe generation integration	GPT-2 model fine-tuned on Recipe1M+; basic pipeline runs end-to-end.
Nov 10 – 16	Illustration and nutrition modules	Stable Diffusion + ControlNet generating step visuals; nutrition PGM integrated.
Nov 17 – 23	Refinement and testing	Optimized inference speed, improved recipe diversity, full prototype ready for demo.

g. Responsible AI Reflection

The cAluldron team prioritizes fairness, transparency, and environmental responsibility in every design decision. Fairness involves ensuring that recipes represent global cuisines rather than a narrow cultural subset, promoting inclusivity for users of all backgrounds. Transparency is achieved by exposing model confidence levels and referencing trusted data sources like USDA FoodData Central. Environmental considerations include optimizing for energy efficiency through model pruning, caching, and quantization to reduce compute usage.

User privacy is another pillar of responsible AI practice. Ingredient photos remain local to the user's machine; no cloud uploads occur without explicit consent. Additionally, all AI-generated outputs are subject to validation rules to prevent unsafe or misleading cooking recommendations.

```
├─ notebooks/                # Jupyter notebooks (core development environment)
│   ├── explore_ingredients/  # Exploratory analysis of ingredient dataset
│   │   ├── explore_ingredient_dataset.ipynb
│   │   └── explore_image_preprocessing.ipynb
│   ├── model_ingredient_recognition/ # CNN-based ingredient recognition
│   │   ├── model_cnn_training.ipynb
│   │   └── model_cnn_inference.ipynb
│   ├── model_recipe_generation/    # Transformer-based recipe generation
│   │   ├── model_transformer_training.ipynb
│   │   └── model_transformer_inference.ipynb
│   ├── model_cooking_refinement/    # RNN for cooking sequence optimization
│   │   ├── model_rnn_training.ipynb
│   │   └── model_rnn_inference.ipynb
│   ├── model_nutrition_estimation/  # Probabilistic models for nutrition
│   │   ├── model_pgm_training.ipynb
│   │   └── model_pgm_inference.ipynb
│   ├── model_illustration_generation/ # Stable Diffusion for line-art illustrations
│   │   ├── model_stable_diffusion_lineart.ipynb
│   │   └── model_stable_diffusion_lineart_improved.ipynb
│   ├── pipeline_recipe_app/        # End-to-end recipe generation pipeline
│   │   ├── pipeline_photo_to_recipes.ipynb
│   │   └── pipeline_interactive_guide.ipynb
│   └── utils_recipe/              # Shared utilities
│       ├── utils_image_processing.ipynb
│       ├── utils_model_loading.ipynb
│       ├── utils_visualization.ipynb
│       ├── utils_reproducibility.ipynb
│       ├── utils_validation.ipynb
│       └── utils_logging.ipynb
├─ data/                        # Data directory
│   ├── raw/                    # Original datasets
│   │   ├── ingredient_images/  # Training images of ingredients
│   │   ├── recipe_corpus/     # Recipe text data
│   │   └── nutrition_database/ # Nutritional reference data (USDA)
│   ├── processed/              # Preprocessed data
│   │   ├── ingredient_features/ # CNN embeddings
│   │   ├── recipe_tokens/      # Tokenized recipes
│   │   └── nutrition_vectors/   # Nutrition embeddings
│   ├── results/                # Model outputs
│   │   ├── generated_recipes/  # AI-generated recipes (JSON)
│   │   ├── illustrations/      # Stable Diffusion-generated line art
│   │   └── nutrition_estimates/ # Calorie and portion calculations
├─ models/                      # Trained model weights
│   ├── cnn_ingredient_recognition.h5
│   ├── transformer_recipe_generation.pt
│   ├── rnn_cooking_refinement.pt
│   ├── pgm_nutrition_estimation.pkl
│   └── stable_diffusion_controlnet/ # Downloaded from HuggingFace
│       ├── stable-diffusion-v1-5/
│       └── control_v11p_sd15_lineart/
```